Locality Regularized Robust-PCRC: A Novel Simultaneous Feature Extraction and Classification Framework for Hyperspectral Images

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***Abstract*— Despite the successful applications of probabilistic collaborative representation classification (PCRC) in pattern classification, it still suffers from two challenges when being applied on hyperspectral images (HSIs) classification: 1) ineffective feature extraction in HSIs under noisy situation; and 2) lack of prior information for HSIs classification. To tackle the first problem existed in PCRC, we impose the sparse representation to PCRC, i.e., to replace the 2-norm with 1-norm for effective feature extraction under noisy condition. In order to utilize the prior information in HSIs, we first introduce the Euclidean distance (ED) between the training samples and the testing samples for the PCRC to improve the performance of PCRC. Then, we bring the coordinate information (CI) of the HSIs into the proposed model, which finally leads to the proposed locality regularized robust PCRC (LRR-PCRC). Experimental results show the proposed LRR-PCRC outperformed PCRC and other state-of-the-art pattern recognition and machine learning algorithms.**

***Index Terms*— Coordinate information (CI), Euclidean distance (ED), hyperspectral image (HSIs), probabilistic collaborative representation classification (PCRC), sparse representation.**

I. Introduction

HYPERSPECTRAL images (HSIs) data has high spectral resolution due to the large spectral range and thus facilitate the discriminative performance on different land cover types [1]. The supervised classification [2] is one of the main topics in HSIs processing. In the supervised classification scenario, the output of testing samples is determined by the given training samples labeled for each class [1]. The great challenges, however, posed by adopting supervised classification on HSIs are due to various causes. First, the ratio of the large number of spectral bands with respect to the limited samples of training pixels is high, i.e., the well-known Hughes phenomenon [3]. Second, the materials of the same category may appear different spectral features and the different classes of samples may share the same spectral characteristic due to the sensory or environmental noises [4], [5].

In order to address the Hughes phenomenon in HSIs or remote sensing images, many advanced supervised classification methods have been proposed and achieved good performance, such as the support vector machine (SVM) [5], extreme learning machine (ELM) [6], [7], sparse multinomial logistic regression (SMLR) [8], [9], etc. In addition, some recent literatures have also been reported for solving the Hughes Phenomenon. For examples, in [10] and [11], the band selection techniques have been proposed for reducing the spectral bands via adaptive subspace partition strategy and optimal clustering framework, respectively; a subpixel component analysis and recurrent attention technique have been proposed for HSI and scene classification, respectively, in [12] and [13]; in [14] and [15], the performance of HSIs classification has been improved by multilabel sample augmentation-based and superpixel-based semisupervised active learning. The above methods have achieved relative good performances. On the other hand, methods such as sparse representation classification (SRC) [16], [17] and collaborative representation classification (CRC) [1] also achieved good performance in HSIs classifications. Different from SVM, ELM, or SMLR, the SRC and CRC do not need the training process because they aim to represent a testing sample directly by the linear combination of all the training samples from all classes with 1-norm and 2-norm on the representation coefficient, respectively [18], [19]. Both methods finally assign the class labels to the corresponding testing samples directly via evaluating the minimum redundancy representation among all the classes [1]. This advantage of SRC and CRC has drawn vast research attentions. However, the SRC has its drawback in HSIs processing. The SRC is time-consuming

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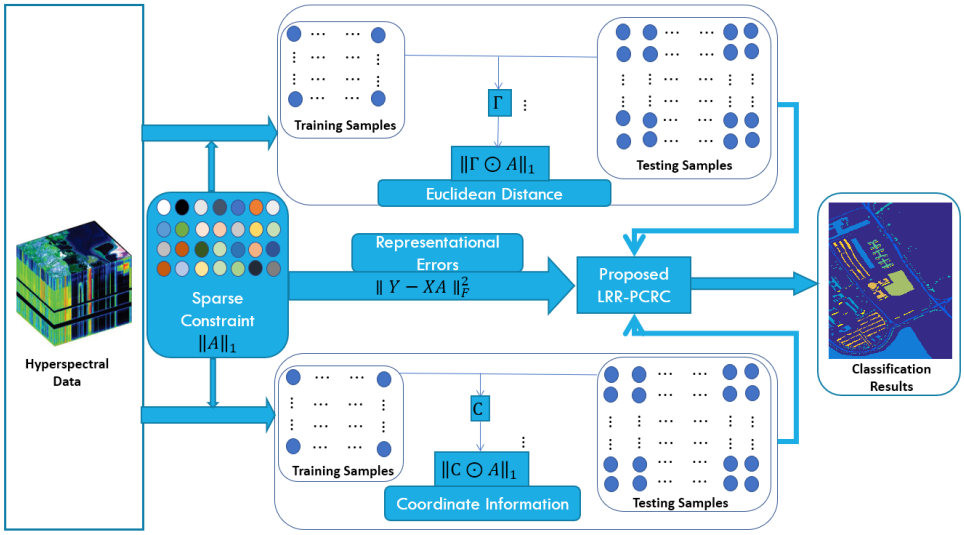


Fig. 1. Classification flowchart of proposed LRR-PCRC algorithm.

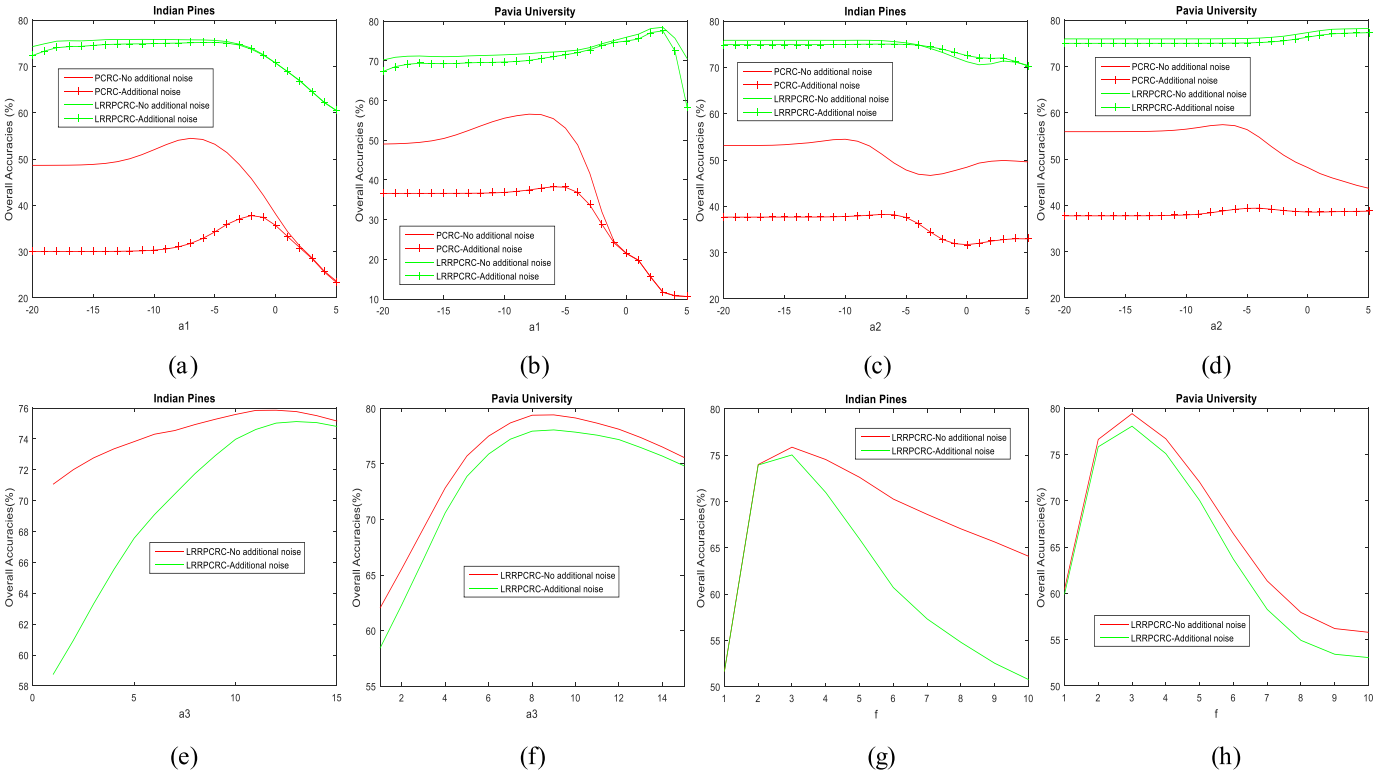


Fig. 2. Impact of PCRC and proposed LRR-PCRC without and with additional noise in Indian Pines and Pavia University. (a) and (b)$\lambda/a1$, (c) and (d) $\beta/a2$, (e) and (f) $\gamma/a3$, and (g) and (h) f.

as the computation of the sparsity 1-norm is a complicated procedure [1]. Meanwhile, Zhang *et al.* [19] pointed out that it is unnecessary to regularize the 1-norm coefficients if the feature dimension of samples is high enough [1]. They argued that the success of SRC can be largely attributed to the collaborative representation between the test sample and the training samples among all classes [19]. Therefore, CRC has attracted many attentions due to its good performance. For example, in [20], a relaxed CRC method has been proposed for multiple features fusion and classification with good performance. In [21], the good classification performance has been produced based on a collaborative neighbor representation method. In [22], a probabilistic collaborative representation classifier (PCRC) for pattern classification based on the viewpoint of probabilities and achieved better performance than CRC and some other methods [22]. Even for HSIs classification, the CRC also produced good performance. For example, a joint within-class collaborative representation has been proposed for HSI classification in [23]. In [17], a nonlocal joint collaborative

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TABLE I
ABBREVIATIONS OF THIS ARTICLE

Abbreviation List			
PCRC	probabilistic collaborative representation classification	SSA	singular spectrum analysis
HSIs	hyperspectral images	LRR-PCRC	LRR-probabilistic collaborative representation classification
ED	Euclidean distance	IALM	inexact augmented Lagrange multiplier
CI	coordinate information	SVM-CK	SVM with composite kernel
LRR	locality regularized robust	AP	attribute profile
SVM	support vector machine	KPI	key performance indicators
ELM	extreme learning machine	OA	overall accuracies
SMLR	sparse multinomial logistic regression	AA	average accuracies
SRC	sparse representation classification	k	kappa coefficient
CRC	collaborative representation classification	CA	category accuracies
GRLD	graph-regularized low rank destriping	Tr	training time
PCA-WS	principal component analysis and wavelet shrinkage	Ts	testing time
Val	Validation	RF	Random Forest
SSAHTV	spectral-spatial adaptive hyperspectral total variation	RoRF-KPCA	rotation random forest-kernel principal component analysis
LRR	Locality Regularized Robust	RoF	Rotation Forest

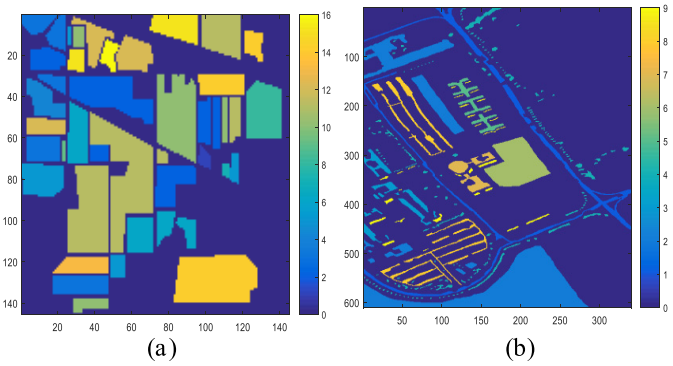


Fig. 3. (a) Ground truth for Indian Pines hyperspectral data set. (b) Ground truth for Pavia University hyperspectral data set.

representation with locally adaptive dictionary has been proposed for HSIs classification. In [24], a joint collaborative representation with multitask learning has been proposed for HSI classification. These CRC-based methods have achieved good performance to some extent, but they ignored a critical issue of CRC that 2-norm is sensitive to noise when HSIs are prone to be corrupted by noise as stated in literature [16].

Noises may be introduced to HSIs during the process of HSIs data acquisition and transmission [25]. Hence, it is important to design an algorithm to extract features from noisy HSIs efficiently and effectively. For this purpose, many algorithms have been proposed, such as graph-regularized low-rank destriping (GRLD) [26], principal component analysis and wavelet shrinkage (PCA-WS) [27], spectral–spatial adaptive hyperspectral total variation (SSAHTV) [28], and singular spectrum analysis (SSA) [25]. However, these methods focused on either feature extraction or denoising independently, resulting in the classification results may not be optimal due to the lack of comprehensive consideration for both preprocessing and classification tasks.

To address the above challenges, we propose a locality regularized robust probabilistic collaborative representation classification (LRR-PCRC) framework to process the feature extraction and classification simultaneously for HSIs classifications. First, we introduce the sparse representation (SR) to the PCRC. That is because it is well-known that the 1-norm SR [29], [30] is robust to characterize the loss function if the data sets are corroded by noises. Then, we introduce the rich prior information of HSIs to PCRC in order to extract the efficient feature from HSIs, including the Euclidean distance (ED) between training samples and testing samples, and the coordinate information (CI) of the HSIs. The motivations can be summarized in two aspects. On one hand, the same category shares similar spectral features to some extent even when the data points are corrupted, hence, the features can be extracted. More importantly, on the other hand, the neighborhood pixels within the regions are more likely to be in the same category [1]. Therefore, the CI within HSIs can be used for extracting the spatial information since the coordinate of the pixels in HSIs cannot be changed even the data points in HSIs are corrupted. For the ED and CI information, we calculate the ED and coordinate distance between each training sample and the whole testing samples and add the prior information to impose the constraints to the representation coefficient. Finally, the locality regularization will be employed for robust HSIs feature extraction and classification simultaneously.

The main contributions of our proposed work can be summarized as follows.

1) A novel LRR-PCRC framework for simultaneous feature extraction and classification is proposed.

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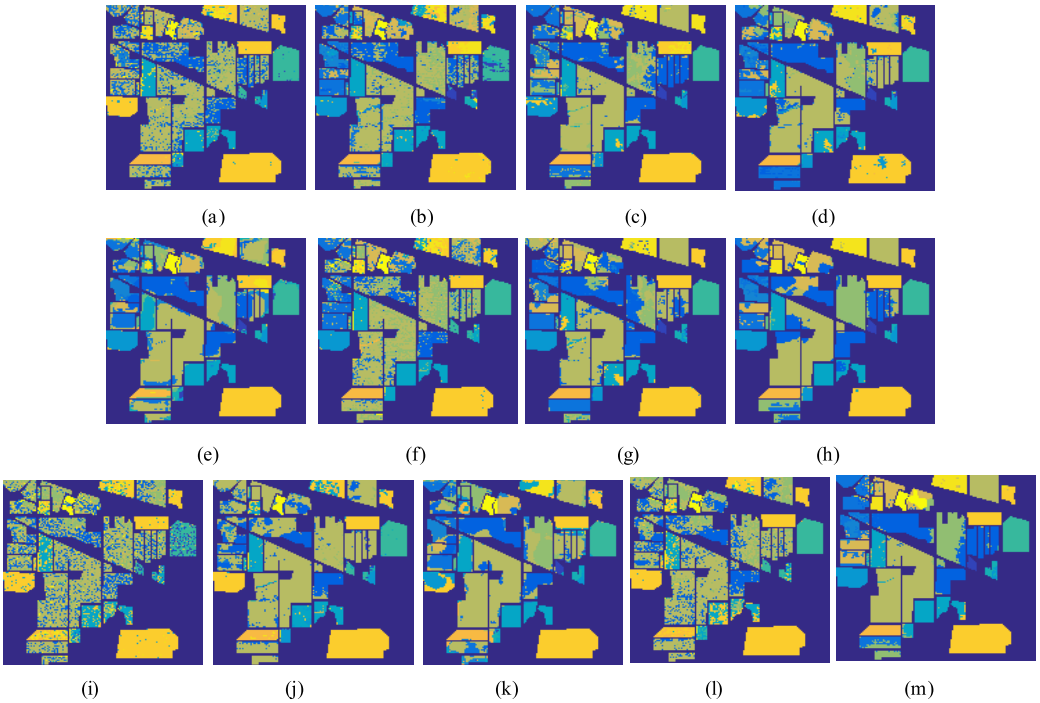


Fig. 4. Classification map of different classifiers on Indian Pines data set.(a)–(h) Without additional noise.(i)–(m) With additional noise (level of noise: $\sigma = 0.02$). (a) and (i) SMRL. (b) and (j) SVM. (c) SMLR-AP. (d) SVM-AP. (e) and (k) SVM-CK. (f) and (l)PCRC. (g) PCRC-AP. (h) and (m) LRR-PCRC.

TABLE II
PARAMETERS SETTING OF PCRC AND LRR-PCRC FOR PARAMETER ANALYSIS

Dataset		Methods	Experiment			
			1	2	3	4
Indian Pines	No additional noise	PCRC	$\beta = 2^{-10}$	$\lambda = 2^{-7}$	-	-
		LRR-PCRC	$\beta = 2^{-10}$	$\lambda = 2^{-10}$	$\lambda = 2^{-10}$	$\lambda = 2^{-10}$
			$f = 3$	$f = 3$	$\beta = 2^{-8}$	$\beta = 2^{-8}$
			$\gamma = 2^{12}$	$\gamma = 2^{12}$	$f = 3$	$\gamma = 2^{12}$
	Additional noise	PCRC	$\beta = 2^{-10}$	$\lambda = 2^{-2}$	-	-
		LRR-PCRC	$\beta = 2^{-10}$	$\lambda = 2^{-10}$	$\lambda = 2^{-10}$	$\lambda = 2^{-10}$
			$f = 3$	$f = 3$	$\beta = 2^{-8}$	$\beta = 2^{-8}$
			$\gamma = 2^{12}$	$\gamma = 2^{12}$	$f = 3$	$\gamma = 2^{12}$
Pavia University	No additional noise	PCRC	$\beta = 2^{-10}$	$\lambda = 2^{-7}$	-	-
		LRR-PCRC	$\beta = 2^{-10}$	$\lambda = 2^{0}$	$\lambda = 2^{0}$	$\lambda = 2^{0}$
			$f = 3$	$f = 3$	$\beta = 2^{3}$	$\beta = 2^{3}$
			$\gamma = 2^{12}$	$\gamma = 2^{12}$	$f = 3$	$\gamma = 2^{9}$
	Additional noise	PCRC	$\beta = 2^{-10}$	$\lambda = 2^{-4}$	-	-
		LRR-PCRC	$\beta = 2^{-10}$	$\lambda = 2^{0}$	$\lambda = 2^{0}$	$\lambda = 2^{0}$
			$f = 3$	$f = 3$	$\beta = 2^{3}$	$\beta = 2^{3}$
			$\gamma = 2^{12}$	$\gamma = 2^{12}$	$f = 3$	$\gamma = 2^{9}$

2) We introduce the sparse constraint to PCRC to tackle the challenge that PCRC is sensitive to the noise.
3) The prior information of ED information and CI between training samples and testing samples is imposed to the PCRC to improve the classification results.

The remaining of this work is structured as follows. Section II introduces the related works. The details of the proposed LRR-PCRC framework are discussed in Section III. Section IV shows the experimental results and the comprehensive analysis. Section V concludes this article with insights of future work.

II. PROBABILISTIC COLLABORATIVE REPRESENTATION CLASSIFICATION (PCRC)

Given N training samples from K classes of an HSI: $X = [X_{N_1}, X_{N_2}, \ldots, X_{N_K}] \in R^{d\times N}$, where d is the number of band in an HSI and X_{N_k} is the data matrix of the kth class training samples ($N_1 + N_2 + \ldots + N_K = N$). Let la_X and S denote the label set of all the classes in X and the linear subspace collaboratively spanned by the total samples in X, respectively [22]. Then given a data point $x \in R^{d\times 1}$ in the collaborative subspace S, it can be represented as follows: $x = X\alpha$, where $\alpha = [\alpha_{N_1}, \alpha_{N_2}, \ldots, \alpha_{N_K}] \in R^{N\times 1}$ is the representation vector.

The PCRC [22] formulated S as a probabilistic collaborative subspace, as it argued that there should be different probabilities of $la_x \in la_X$ for a data point x. Once the 2-norm of α_{N_k} is small, the probability of $P(la_x \in la_{X_{N_k}})$ should be high [22]. It is intuitive to use a Gaussian function to define the probability, that is,

$$P(la_x \in la_{X_{N_k}}) \propto \exp\left(-b\|\alpha_{N_k}\|_2^2\right) \tag{1}$$

where b is a positive constant. Then the probability of every testing sample y that has the sample label of x can be formulated as $P(la_y = la_x)$. Hence, the probabilities that the testing sample y lies in the subspace S can be expressed as

$$P\left(la_y \in la_X\right) = P\left(la_y = la_x | la_x \in la_X\right) P(la_x \in la_X) \tag{2}$$

where $P\left(la_y = la_x | la_x \in la_X\right) \propto \exp(-c\|y - x\|_2^2)$, c is a positive constant. Then (2) can be rewritten as

$$\begin{aligned} P\left(la_y \in la_X\right) &\propto \exp\left(-\left(c\|y - X\alpha\|_2^2 + d\|\alpha\|_2^2\right)\right) \\ &= \exp\left(-\left(\|y - X\alpha\|_2^2 + \lambda\|\alpha\|_2^2\right)\right) \end{aligned} \tag{3}$$

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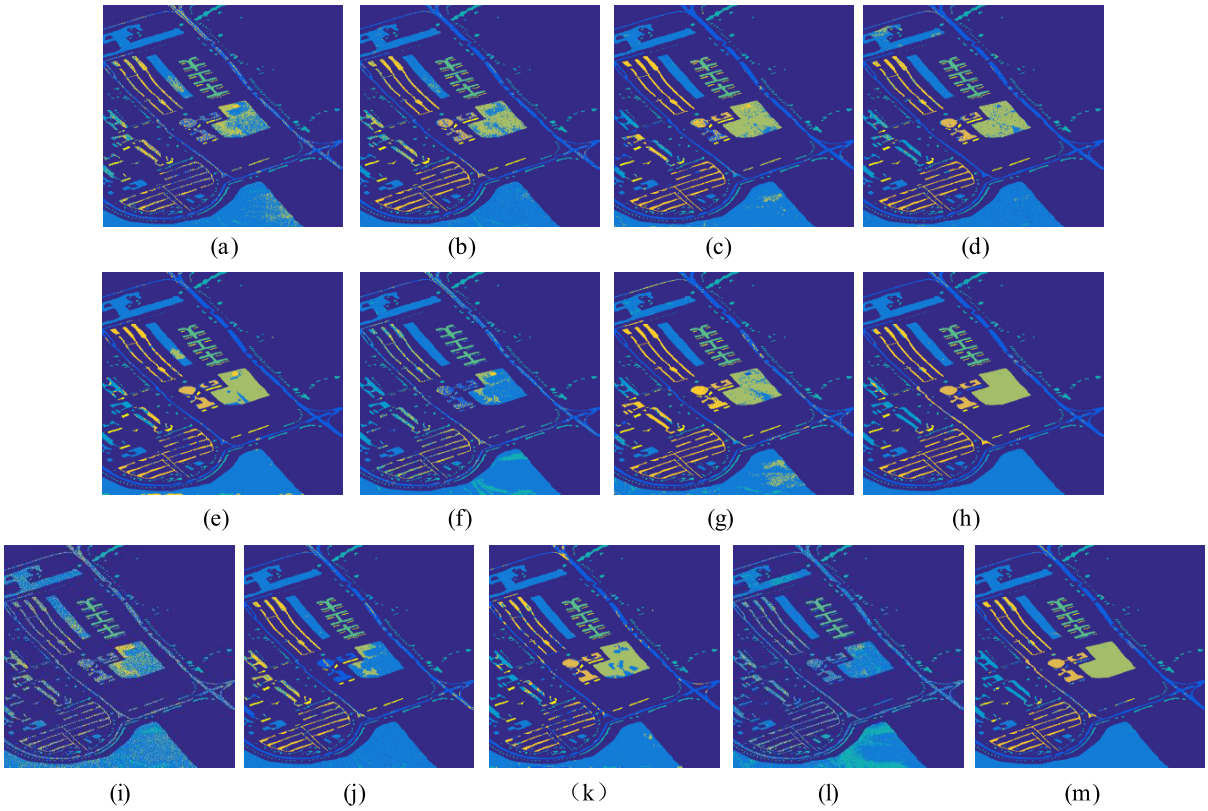


Fig. 5. Classification map of different classifiers on Pavia University data set. (a)–(h) Without additional noise. (i)–(m) With additional noise (level of noise: $\sigma = 0.02$). (a) and (i) SMRL. (b) and (j) SVM. (c) SMLR-AP. (d) SVM-AP. (e) and (k) SVM-CK. (f) and (l) PCRC. (g) PCRC-AP. (h) and (m) LRR-PCRC.

TABLE III
PARAMETERS SETTING OF THE CONTRIBUTIONS FOR CONTRIBUTIONS ANALYSIS

	Indian Pines			Pavia University		
	LRR-PCRC			LRR-PCRC		
	SR	SR+ED	SR+ED+CI	SR	SR+ED	SR+ED+CI
No noise	$\lambda = 2^{-6}$	$\lambda = 2^{-9}$	$\tau^0 = 2^{-10}; \beta = 2^{-8}$	$\lambda = 2^{-10}$	$\lambda = 2^{-7}$	$\tau^0 = 2^{-10}; \beta = 2^{-6}$
	$\beta = 2^{-15}$	$\beta = 2^{-13}$	$\gamma = 2^{12}; f = 3$	$\beta = 2^{-15}$	$\beta = 2^{-13}$	$\gamma = 2^{5}; f = 3$
Noise	$\lambda = 2^{-4}$	$\lambda = 2^{-4}$	$\tau^0 = 2^{-10}; \beta = 2^{-6}$	$\lambda = 2^{-7}$	$\lambda = 2^{-4}$	$\tau^0 = 2^{-10}; \beta = 2^{-4}$
	$\beta = 2^{-9}$	$\beta = 2^{-8}$	$\gamma = 2^{12}; f = 3$	$\beta = 2^{-20}$	$\beta = 2^{-13}$	$\gamma = 2^{13}; f = 3$

where $\lambda = d/c$ is a constant. Since $x = X\alpha = \sum_{k=1}^{K} X_{N_k}\alpha_{N_k} = \sum_{k=1}^{K} x_k$, where x_k is equal to $X_{N_k}\alpha_{N_k}$ and can be seen as a sample that belongs to the kth class, then the probability that the testing sample y belongs to kth class can be formulated as

$$\begin{aligned} P(la_y = k) &= P(la_y = la_x | la_x = k) P(la_x = k) \\ &= P(la_y = la_x | la_x = k) P(la_y = k | la_x \in la_X) \\ &\quad P(la_x \in la_X) \end{aligned} \tag{4}$$

where $P(la_x = k | la_x \in la_X)$ can be formulated as

$$P(la_x = k | la_x \in la_X) \propto \exp\left(-\beta \|x - x_k\|_2^2\right) \tag{5}$$

where β is a positive constant.

Since $P(la_y = la_x | la_x \in la_X)$ is independent of the kth class, once k belongs to la_X, i.e., $P(la_y = la_x | la_x \in la_X) = P(la_y = la_x | la_x = k) \propto \exp\left(-c\|y - x\|_2^2\right)$, then $P(la_y = k)$ can be expressed as

$$\begin{aligned} &P(la_y = k) \\ &= P(la_y \in la_X) P(la_x = k | la_x \in la_X) P(la_y = la_x | la_x = k) \\ &\propto \exp\left(-\left(\|y - x\|_2^2\right) + \lambda\|\alpha\|_2^2 + \beta\|x - x_k\|_2^2\right) \\ &= \exp\left(-\left(\|y - X\alpha\|_2^2\right) + \lambda\|\alpha\|_2^2 + \beta\|X\alpha - X_{N_k}\alpha_{N_k}\|_2^2\right). \end{aligned} \tag{6}$$

Assuming the probabilities of $P(la_y = k)$ are independent, then all kinds of probabilities ($k = 1, 2, \ldots, K$) can be computed by

$$\begin{aligned} &\max P(la_y = 1, la_y = 2, \ldots, la_y = K) \\ &= \max \prod_k P(la_y = k) \\ &\propto \exp - \left(\|y - X\alpha\|_2^2 + \lambda\|\alpha\|_2^2 \right. \\ &\qquad \left. + \beta/K \sum_{k=1}^{K} \|X\alpha - X_{N_k}\alpha_{N_k}\|_2^2 \right). \end{aligned} \tag{7}$$

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TABLE IV

CLASSIFICATION ACCURACY (%) AND STANDARD DEVIATION WITH 20 TRAINING SAMPLES PER CLASS FOR INDIAN PINES AND PAVIA UNIVERSITY DATA SET (BEST RESULTS IN BOLD)

	Index	Indian Pines					Pavia University				
		PCRC	LRR-PCRC				PCRC	LRR-PCRC			
			SR	SR+ED	SR +CI	SR+CI+ED		SR	SR+ED	SR +CI	SR+CI+ED
No noise	OA	61.2±2.8	68.0±2.7	71.5±2.4	**91.0±0.7**	**91.0±0.8**	62.0± 2.8	66.1±4.9	77.1± 2.3	86.8±3.2	**90.4±1.2**
	AA	72.4±1.9	70.8±2.5	83.6±0.9	93.9±0.7	**94.1±0.5**	68.8± 2.1	64.0± 2.7	83.0±0.8	88.7±1.7	**93.5±0.6**
	k	56.8±2.9	63.7±3.1	68.1±2.6	**89.8± 0.8**	89.7±0.9	51.4± 3.0	56.1±5.1	70.7±2.6	83.0±4.0	**87.5± 1.5**
Noise 0.02	OA	43.7±1.3	51.7±1.4	66.7±2.9	87.5±0.7	**87.6±0.8**	39.4± 3.4	44.1±6.0	65.1±2.7	84.5±2.1	**88.7 ± 1.3**
	AA	45.1±0.9	49.6± 1.5	77.6± 1.7	91.0±1.1	**91.8±0.7**	49.0± 1.0	44.4±1.7	74.1±0.9	83.6± 1.1	**89.7 ± 0.7**
	k	37.2±1.2	45.2±1.6	62.8±3.0	85.8±0.7	**85.9±0.9**	28.4 ±2.6	32.4± 5.1	56.4±2.9	80.0± 2.6	**85.3 ± 1.6**

TABLE V

CLASSIFICATION ACCURACY (%) AND STANDARD DEVIATION IN INDIAN PINES

	Index	No noise		Noise 0.02		Noise 0.04		Noise 0.06		Noise 0.08	
		PRRC	LRR-PCRC	PRRC	LRR-PCRC	PRRC	LRR-PCRC	PRRC	LRR-PCRC	PRRC	LRR-PCRC
20	OA	61.2±2.8	91.0±0.8	43.7±1.3	87.6±0.8	30.4±0.9	83.5±1.1	23.6±0.9	80.1± 1.4	19.8± 0.9	77.1± 1.7
	AA	72.4±1.9	94.1±0.5	45.1±0.9	91.8±0.7	32.6±1.3	87.3±1.1	25.8±1.3	82.6± 1.3	21.8±1.0	78.1± 0.9
	k	56.8±2.9	89.7±0.9	37.2±1.2	85.9±0.9	23.2±0.9	81.3±1.3	16.3± 0.8	77.5± 1.6	12.5± 0.7	74.3± 1.8
25	OA	63.4±2.5	91.9±1.1	45.4± 1.4	88.8±0.9	31.6± 1.1	84.7±1.0	25.1± 1.0	81.3±1.0	21.6±0.9	78.5±0.8
	AA	73.3±2.3	95.0±0.4	46.0±1.0	92.4±0.9	33.8± 0.8	87.2±1.3	27.0±0.7	81.9±1.7	23.1± 0.5	77.7±1.3
	k	59.2±2.5	90.8±1.2	39.0±1.2	87.2± 1.0	24.4± 0.9	82.6±1.1	17.7±0.7	78.8±1.1	14.2±0.6	75.7±0.9
30	OA	63.9±2.4	92.7±0.9	45.5± 0.9	89.59±0.6	31.8 ± 1.1	85.7±0.6	25.4±1.2	82.5±0.8	22.2±1.1	79.6±1.0
	AA	73.2±1.3	95.1± 0.8	45.8± 0.9	92.7±0.6	33.7 ± 0.9	87.4±1.7	27.4±1.1	82.3±1.4	24.0± 0.9	78.7±1.0
	k	59.7±2.4	91.7± 1.0	39.2±0.9	88.1±0.7	24.6 ± 1.0	83.7±0.7	18.0±1.0	80.1±0.9	14.8±0.9	76.9±1.1

TABLE VI

CLASSIFICATION ACCURACY (%) AND STANDARD DEVIATION WITH DIFFERENT TRAINING NUMBERS AND NOISE LEVELS IN PAVIA UNIVERSITY DATA SET

	Index	No noise		Noise 0.02		Noise 0.04		Noise 0.06		Noise 0.08	
		PRRC	LRR-PCRC	PRRC	LRR-PCRC	PRRC	LRR-PCRC	PRRC	LRR-PCRC	PRRC	LRR-PCRC
20	OA	62.0±2.8	90.4±1.2	39.4± 3.4	88.7 ± 1.3	33.5± 2.7	84.1±1.4	31.4± 2.5	79.4±1.7	30.2± 2.5	73.8±2.1
	AA	68.8±2.1	93.5±0.6	49.0± 1.0	89.7 ± 0.7	42.0± 1.1	80.9±0.4	38.9± 1.1	78.1±0.8	37.2± 1.1	74.7±1.2
	k	51.4±3.0	87.5±1.5	28.4 ±2.6	85.3 ± 1.6	22.3± 2.0	79.5±1.7	20.1± 1.9	73.8±2.0	18.9± 1.8	67.2±2.3
25	OA	63.4±1.7	91.6±1.7	44.2± 2.7	90.0± 2.0	37.7± 2.3	85.2±1.9	34.6± 1.9	81.1±1.7	32.8±1.6	76.0± 2.1
	AA	69.9±1.8	94.4±0.5	51.0± 1.2	91.0± 0.8	44.2± 1.0	81.0± 0.8	40.7± 1.0	78.4±0.9	38.7± 0.8	75.0±1.1
	k	53.0±1.7	89.1± 2.1	32.5± 2.3	87.0±2.5	26.0± 1.7	80.9± 2.4	23.0± 1.4	75.7±2.1	21.2± 1.2	69.7±2.4
30	OA	64.0±1.2	92.4±1.1	45.0± 3.0	90.4±1.5	39.5± 2.6	85.4±1.9	36.8± 2.4	80.8±2.3	34.9± 2.1	75.2±2.4
	AA	71.2±1.0	94.8±0.5	52.5± 1.0	91.3±0.7	45.4± 0.8	82.2±0.8	41.6± 0.8	79.6±1.4	39.4± 0.8	76.5±1.5
	k	53.8±1.4	90.1± 1.5	33.3± 2.6	87.5±1.9	27.4± 2.1	81.1±2.3	24.7± 2.0	75.5±2.8	22.8± 1.7	68.9± 2.7

Then ignoring the constant term after applying the logarithmic operator to the above equation, we can have

$$\hat{\alpha} = \arg\min_{\alpha}\left\{ \|y - X\alpha\|_2^2 + \lambda\|\alpha\|_2^2 + {}^{\beta}\!/_{K}\sum_{k=1}^{K}\|X\alpha - X_{N_k}\alpha_{N_k}\|_2^2 \right). \quad (8)$$

The solution of (8) can be expressed as

$$\hat{\alpha} = \left(X^T X + \lambda I + {}^{\beta}\!/_{K}\sum_{k=1}^{K}\bar{\bar{X}}_{N_k}^T\bar{\bar{X}}_{N_k} \right)^{-1} X^T y \quad (9)$$

where $\bar{\bar{X}}_{N_k} = X - \bar{X}_{N_k}$, $\bar{X}_{N_k} = [0, 0, \ldots, X_{N_k}, \ldots 0] \in R^{d\times N}$.

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TABLE VII
TRAINING/TESTING SAMPLES IN THE TWO DATA SETS

Indian Pines			Pavia University		
Index/category	Train	Test	Index/Category	Train	Test
1 Alfalfa	3	51	1 Asphalt	66	6565
2 Corn-no till	14	1420	2 Meadows	186	18463
3 Corn-min till	8	826	3 Gravel	20	2079
4 Corn	4	230	4 Trees	30	3034
5 Grass/pasture	5	492	5 Metal sheets	13	1332
6 Grass/tree	8	739	6 Bare soil	50	4979
7 Grass/pasture-mowed	3	23	7 Bitumen	13	1317
8 Hay-windrowed	5	484	8 Bricks	37	3645
9 Oats	2	18	9 Shadows	10	937
10 Soybeans-no till	10	958			
11 Soybeans-min till	24	2444			
12 Soybeans-clean till	7	607			
13 Wheat	4	208			
14 Woods	13	1281			
15 Bldg-grass-tree-drives	5	375			
16 Stone-steel towers	4	91			

Finally, the label of testing sample y can be formulated as

$$la_y = \arg\max_k \exp\left(-\|X\hat{\alpha} - X_{N_k}\hat{\alpha}_{N_k}\|_2^2\right). \quad (10)$$

III. PROPOSED FRAMEWORK

A. LRR-PCRC Model

Although the PCRC has achieved good performance, there still exist some problems. As mentioned above, PCRC is sensitive to noises. The performance of PCRC will degrade if HSIs contain much noise. In order to address this drawback, we propose the LRR-PCRC for HSIs feature extraction and classification simultaneously. For N training samples from K classes of an HSI $X = [X_{N_1}, X_{N_2}, \ldots, X_{N_K}] \in R^{d\times N}$ and n testing samples $Y = [Y_1^*, Y_2^*, \ldots, Y_n^*] \in R^{d\times n}$, we need to represent the n testing samples Y using N training samples and assume both of them are corrupted by noise, that is,

$$\begin{aligned} Y + E_1 &= (X + E_2)A = XA + E_2A \\ &\xrightarrow{\Delta} Y = XA + (E_2A - E_1) \xrightarrow{\Delta} Y = XA + E \end{aligned} \quad (11)$$

where $E_1 \in R^{d\times n}$, $E_2 \in R^{d\times N}$, and $E \in R^{d\times N}$ are noise, $A \in R^{N\times n}$ is the representation coefficient. From (11), we can see that the coefficient A needs to represent the n testing samples Y using N training samples under the situation that both of them are corrupted. In order to address this problem, we can formulate the PCRC to the following structure:

$$\hat{A} = \arg\min_A \left\{ 1/2\|Y - XA\|_F^2 + \lambda\|A\|_1 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k}A_{N_k,:}\|_F^2 \right\} \quad (12)$$

In (12), we replace the 2-norm with 1-norm considering that the 1-norm is robust to noise [30]. Furthermore, the feature extraction performance can be improved if we incorporate the rich prior information in HSIs to the model, thus producing better classification results. Recall the ED between training samples and testing samples, the spectral signal will be similar if the training samples and testing samples belong to the same class. Hence, we incorporate this information in (12), and then the model can be rewritten as

$$\hat{A} = \arg\min_A \left\{ 1/2\|Y - XA\|_F^2 + \lambda\|\Gamma \odot A\|_1 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k}A_{N_k,:}\|_F^2 \right\} \quad (13)$$

where $\odot$ is Hadamard operator [24], $\Gamma \in R^{N\times n}$ and $\Gamma_{i,j} = \left\|Y_j^* - X_i\right\|_2^2$ $(i = 1, 2, \ldots, N; j = 1, 2, \ldots, n)$. Recall the coefficient A, each column of A, $A_{:,j} \in R^{N\times 1}$, represents the similarity between the testing pixel Y_j and the whole training samples X. The probability of the testing pixel Y_j belonging to the kth class should be higher when the 2-norm of coefficient $A_{N_k,j}$ $((\sum_{Ni=1}^{N_k}\left|A_{Ni,j}\right|^2)^{1/2})$ is smaller which has been explained in (1). Also, we can deduce that the probability of the testing pixel Y_j belonging to the kth class should be higher when the 1-norm of coefficient $A_{N_k,j}$ $((\sum_{Ni=1}^{N_k}\left|A_{Ni,j}\right|)^{1/2})$ is smaller since the 1-norm of coefficient $A_{N_k,j}$ is similar to the 2-norm of coefficient $A_{N_k,j}$. On the other hand, the 1-norm sparse representation is robust to noise as reported in [30].

Furthermore, we use the ED between the testing samples and training samples Γ to enhance the coefficient A, which can improve the classification performance both in terms of accuracies and robustness. Other than the ED between the testing samples and the training samples, the CI of training samples and testing samples can also enhance the coefficient A. First, we calculate the coordinate distances between each training sample and the whole testing samples, then the CI which contains the locality information of training samples and testing samples will be used for enhancing the performance of HSIs classification. This can be explained from two aspects: first, the probabilities that two samples belonging to the same class will be larger if they are located closely [4]; second, even the data points in HSIs are corrupted by noise, the CI of each pixel in HSIs remains the same. Hence, (13) can be represented as

$$\hat{A} = \arg\min_A \left\{ 1/2\|Y - XA\|_F^2 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k}A_{N_k,:}\|_F^2 + \lambda\|\Gamma \odot A\|_1 + \gamma\|C \odot A\|_1 \right\} \quad (14)$$

where $C \in R^{N\times n}$ and $C_{i,j} = (CC_{i,j}CC_{i,j})/(\sum_{i=1}^{N} CC_{i,j}CC_{i,j})$, $CC_{i,j}$ can be expressed as follows:

$$CC_{i,j} = \left(\left|h_i - h_j\right| + \left|w_i - w_j\right|\right)^f \quad (15)$$

where h_i and w_i are the abscissa and ordinate of sample in the image, respectively. f is the smooth parameter that adjusts the distance decay speed. Fig. 1 shows the flowchart of the proposed LRR-PCRC.

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TABLE VIII
CLASSIFICATION RESULTS IN INDIAN PINES DATA SET (1% TRAINING SAMPLES) (BEST RESULTS IN BOLD)

	No additional noise								Noise level $\sigma = 0.02$				
NO.	SMLR	SVM	SMLR-AP	SVM-AP	SVM-CK	PCRC	PCRC-AP	LRR-PCRC	SMLR	SVM	SVM-CK	PCRC	LRR-PCRC
1	37.2±17.0	67.0± 18.2	92.1±6.6	93.3±4.9	67.0±21.2	21.5±28.5	90.1±3.4	**97.8±2.9**	13.7±7.5	37.4±22.2	74.3±17.0	0.9± 2.1	**94.7±8.9**
2	61.4±7.1	63.3±8.3	70.1±4.5	66.9±5.9	70.4±11.1	63.6±11.4	68.6 ±4.3	**80.7±5.0**	34.4±5.0	39.5±5.1	64.0±8.8	30.5± 9.6	**76.1±6.0**
3	25.3±6.8	45.9±10.7	70.6±6.3	**78.9±6.5**	64.9±16.2	35.7±15.5	77.3 ±7.1	67.3±5.1	10.3±3.2	20.6±6.4	48.9±10.1	1.1± 1.1	**60.8±9.8**
4	20.6±9.7	52.3±12.7	55.1±12.7	59.6±14.4	54.9±21.9	24.5±14.3	53.6 ±10.4	**83.9±10.3**	2.6±2.8	26.7±12.4	61.5±12.9	0.3± 0.9	**79.9±16.8**
5	47.6±13.4	72.8±17.0	75.4±13.0	74.8±13.1	71.5± 8.7	57.4±12.1	64.8 ±15.3	**79.2±8.6**	9.7±4.8	41.8±13.6	53.6±8.7	0.4± 1.0	**76.0±7.4**
6	90.5±3.8	80.0±12.0	91.4±3.8	91.2±4.8	86.8± 8.1	**95.3±1.7**	84.7 ± 7.6	82.9±9.8	70.1±10.4	**80.8± 6.8**	85.4± 8.0	64.5± 12.1	80.7±10.9
7	66.0±11.3	86.0±12.0	91.7±8.0	95.2±3.2	93.9± 5.8	31.7±14.0	**99.1 ± 2.7**	95.6±3.5	13.0±7.1	76.0±18.9	**96.0±5.9**	0± 0	93.4±2.2
8	88.0±9.1	80.4±16.6	99.4±0.3	99.3±0.1	88.2± 8.4	98.7±2.1	**99.6 ± 0.1**	**99.6±0.8**	65.7±9.0	75.7±14.7	96.1±1.8	99.1± 0.5	**99.9±0.1**
9	33.8±19.8	82.2± 24.9	58.3±19.8	**97.2±8.7**	86.1± 11.1	32.2±14.0	25.0 ±13.4	96.1± 8.3	0± 0	34.4±29.2	**87.7± 5.7**	0± 0	79.4±22.2
10	34.5±8.6	57.6±13.7	60.3±15.5	59.9±11.1	60.0± 14.4	36.2±20.1	32.5 ± 12.7	**78.3±5.0**	15.5±4.6	31.9±12.6	53.1±6.0	3.7±4.6	**75.4±4.2**
11	69.0±5.5	72.7±4.7	86.9±5.8	76.0±7.7	79.3± 7.2	64.0±10.5	**92.3 ± 4.2**	92.1±2.9	58.9±4.9	74.4±7.8	82.9±5.3	88.5± 3.2	**92.1±3.5**
12	28.6±8.4	44.8±7.0	48.5±14.6	48.8±12.0	46.3± 10.2	40.3±11.3	37.4 ± 9.2	**77.9± 16.3**	9.1±3.6	22.8± 12.4	45.8±8.3	1.5± 2.4	**75.6±17.5**
13	93.6±5.7	87.7±10.1	98.8±0.6	98.8±0.4	93.5+ 9.3	**99.9± 0.2**	99.5 ± 0	99.8± 0.2	46.1±8.9	90.1±6.1	91.2±5.3	40.2± 20.0	**100.0±0**
14	89.4±5.5	86.6±8.6	92.4±6.7	87.5±7.8	87.3± 5.1	97.7±1.3	93.9 ± 5.1	**99.6±0.4**	93.6±3.7	90.3± 8.1	91.8±4.1	99.7± 0.1	**99.6±0.4**
15	44.7±11.8	30.2±8.9	73.3±15.5	70.6±13.9	52.8±9.0	31.7±10.5	82.2 ± 6.7	**90.1±10.9**	16.1±3.8	19.3±7.2	63.9±11.6	4.1± 2.1	**90.1±10.9**
16	82.5±7.8	84.3±2.4	87.9±6.1	**97.4±2.6**	93.9±7.6	96.2±2.1	96.1 ± 2.4	94.9±4.6	76.8±7.6	85.4±3.3	**95.3±6.2**	92.7± 3.4	95.2±4.8
OA	61.2±2.1	67.2±1.2	78.6±1.8	75.7±2.0	73.6± 4.8	64.1± 2.2	76.5±1.8	**86.4± 1.5**	44.3± 1.5	56.2± 1.2	72.1± 1.8	49.5±1.5	**84.4±1.3**
AA	57.1±2.3	68.4±2.3	78.3±1.5	81.0±1.2	74.8± 3.1	57.9± 2.5	74.8 ± 1.0	**88.5± 1.6**	33.5±1.8	52.9± 4.3	74.5 ± 2.1	32.9± 1.9	**85.6±2.0**
k	55.2±2.4	62.6± 1.4	75.5±2.1	72.4±2.2	69.9± 5.5	58.8± 2.4	72.8 ± 2.1	**84.5± 1.7**	35.1± 1.8	49.2±1.6	67.9±2.1	38.9± 1.9	**82.2± 1.6**

B. Solutions for LRR-PCRC

In this section, we will derive the optimization algorithm to solve the proposed LRR-PCRC model based on the inexact augmented Lagrange multiplier (IALM) [31]. Recall that the proposed LRR-PCR has three contributions compared with the PCRC, i.e., SR, ED between the training samples and testing sample, CI of the HSIs. Now we derive the solution for the proposed LRR-PCRC as follows. (The other solutions of the PCRC with SR or ED, PCRC with SR and ED will be shown in Supplementary.)

First, we introduce two auxiliary variables H and J to convert (14) by splitting the variable

$$\min_{A,H,J} \left\{ {}^1\!/_2 \|Y - XA\|_F^2 + {}^\beta\!/_K \sum_{k=1}^{K} \|XA - X_{N_k} A_{N_k,:}\|_F^2 + \lambda \|\Gamma \odot H\|_1 + \gamma \|C \odot J\|_1 \right\} \quad \text{s.t. } H = A, \quad J = A. \tag{16}$$

Then, the corresponding augmented Lagrangian function for (16) can be rewritten as

$$\min_{A,H,J} \left\{ {}^1\!/_2 \|Y - XA\|_F^2 + {}^\beta\!/_K \sum_{k=1}^{K} \|XA - X_{N_k} A_{N_k,:}\|_F^2 \right\} + \lambda \|\Gamma \odot H\|_1 + \gamma \|C \odot J\|_1 + \langle Y_1, H - A\rangle + \langle Y_2, J - A\rangle + {}^\tau\!/_2 \left(\|H - A\|_F^2 + \|J - A\|_F^2 \right) \Big\} \tag{17}$$

where Y_1 and Y_2 are the Lagrange multipliers. Then the alternative optimization algorithm [32] can be applied to solve the model of (17). Details are introduced as follows.

Update H : Fix A and J, then the H can be updated as

$$H^{t+1} = \min_H {}^\lambda\!/_{\tau^t} \left\| \Gamma \odot H^t \right\|_1 + {}^1\!/_2 \left\| H^t - A^t + Y_1^t/\tau^t \right\|_F^2. \tag{18}$$

The solution of (18) can be solved by the transformation of the soft-threshold rule [33]

$$H^{t+1} = \text{soft}\left(A^t - \frac{Y_1^t}{\tau^t}, \Gamma \times {}^\lambda\!/_{\tau^t} \right) = \max\left\{ 0, \text{abs}(e) - \left(\Gamma \times {}^\lambda\!/_{\tau^t} \right) \right\} \times \text{sign}(e) \tag{19}$$

where $e = A^t - Y_1^t/\tau^t$.

Update J: Fix A and H, J can be updated as

$$J^{t+1} = \min_J {}^\gamma\!/_{\tau^t} \|C \odot J\|_1 + {}^1\!/_2 \left\| J^t - A^t + Y_2^t/\tau^t \right\|_F^2. \tag{20}$$

The solution of (20) can be computed as

$$J^{t+1} = \text{soft}\left(A^t - \frac{Y_2^t}{\tau^t}, \Gamma \times {}^\lambda\!/_{\tau^t} \right) = \max\left\{ 0, \text{abs}(e) - \left(\Gamma \times {}^\lambda\!/_{\tau^t} \right) \right\} \times \text{sign}(e) \tag{21}$$

where $e = A^t - Y_2^k/\tau^t$.

Update A: Fix H and J, A and be updated as

$$A^{t+1} = \min_A {}^1\!/_2 \|Y - XA^t\|_F^2 + {}^\beta\!/_K \sum_{k=1}^{K} \|XA^t - X_{N_k} A_{N_k,:}^t\|_F^2 + {}^\tau\!/_2 \left(\|H^t - A^t + Y_1^t/\tau^t\|_F^2 + \left\| J^t - A^t + \frac{Y_2^t}{\tau^t} \right\|_F^2 \right). \tag{22}$$

Then the solution of (22) can be achieved by the first-order derivation

$$A^{t+1} = \left(X^T X + {}^\beta\!/_K \sum_{k=1}^{K} \bar{\bar{X}}_{N_k}^T \bar{\bar{X}}_{N_k}) + 2\tau^t I \right)^{-1} \left(X^T Y + \tau^t \left(H^t + J^t \right) + Y_1^t + Y_2^t \right). \tag{23}$$

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TABLE IX
CLASSIFICATION RESULTS IN PAVIA UNIVERSITY (1% TRAINING SAMPLES) (BEST RESULTS IN BOLD)

	No additional noise								Noise level $\sigma = 0.02$				
No.	SMLR	SVM	SMLR-AP	SVM-AP	SVM-CK	PCRC	PCRC-AP	LRR-PCRC	SMLR	SVM	SVM-CK	PCRC	LRR-PCRC
1	81.2±2.8	89.0±1.8	**94.2±1.7**	94.0±1.4	93.5±2.1	71.5±4.6	77.2±4.6	88.6±1.9	52.9±4.8	89.5±2.6	90.2±2.3	28.8±5.2	**86.8±2.0**
2	93.1±1.3	97.3±0.6	96.7±0.7	97.6±0.6	98.2±0.7	90.0±1.6	93.8±1.8	**99.3±0.6**	80.8±3.0	95.5±3.0	97.3±0.7	68.9±7.8	**99.1±0.7**
3	51.5±6.5	62.9±7.9	58.0±9.4	81.1±8.8	79.8±4.3	75.2±13.6	36.0±10.3	**96.2±4.1**	19.3±4.3	19.5±12.7	72.8±4.2	15.1±6.3	**94.8±4.9**
4	81.7±4.8	87.1±3.6	90.7±3.3	93.5±2.2	92.0±2.7	97.7±0.9	**98.0±1.2**	82.8±2.5	69.9±3.1	76.1±4.7	91.4±2.7	**96.0±3.7**	82.1±3.2
5	98.4±1.2	98.9±0.5	98.1±2.0	99.5±0.3	99.8±0.1	**100.0±0**	99.8±0.1	99.3±0.1	95.7±2.6	97.4±5.0	99.6±0.6	**99.9±0**	99.4±0.2
6	47.2±2.6	75.3±2.5	83.7±1.8	87.2±3.1	92.8±1.9	23.0±4.6	75.4±6.3	**99.4±0.3**	31.2±3.9	30.7±9.6	82.4±3.4	10.1±5.0	**99.4±0.3**
7	24.1±6.8	75.9±7.3	51.6±6.3	86.3±3.1	86.7±4.5	5.0±3.2	11.0±6.8	**99.3±0.5**	8.7±2.8	14.8±19.8	72.6±7.5	7.1±2.3	**99.1±0.7**
8	65.5±4.8	84.8±3.0	93.6±2.5	93.2±1.4	82.7±4.2	52.6±17.6	**95.7±2.2**	94.2±2.7	33.3±5.8	84.8±7.0	73.1±3.0	20.7±6.0	**94.5±2.2**
9	59.6±12.1	99.8±0.1	84.2±7.9	99.8±0.1	93.8±2.9	16.3±11.8	**99.8±0.1**	96.9±4.8	14.6±4.5	**99.7±0.3**	94.0±2.1	1.4±1.1	57.3±7.3
OA	77.9±0.7	89.4±0.4	90.6±0.4	94.1±0.5	93.8±0.4	71.9±0.4	84.4±1.1	**95.8±0.4**	59.5±1.0	78.6±0.8	90.0±0.4	48.5±3.7	**94.5±0.4**
AA	66.9±1.4	85.7±1.1	83.4±1.4	92.5±0.9	91.0±0.8	59.0±1.8	76.3±1.6	**95.1±0.9**	45.2±1.2	67.6±2.3	85.9±0.8	38.7±1.8	**90.3±1.0**
k	70.0±0.9	85.8±0.6	87.4±0.5	92.2±0.7	91.7±0.5	61.7±0.7	79.2±1.6	**94.4±0.6**	45.7±1.3	70.4±1.1	86.6±0.6	28.8±5.0	**92.6±0.5**

TABLE X
TRAINING, VALIDATION AND TESTING RESULTS IN INDIAN PINES DATA SET (FIVE SAMPLES PER CLASS FOR TRAINING, FIVE SAMPLES PER CLASS FOR VALIDATION, REMAINING FOR TESTING)

	SMLR			SVM			SVM-CK			PCRC			LRR-PCRC		
NO.	Tr	Val	Ts	Tr	Val	Ts	Tr	Val	Ts	Tr	Val	Ts	Tr	Val	Ts
1	100±0	73.0±12.5	70.0±3.2	100±0	84.0±10.7	81.4±6.6	99.0±3.1	79.0±15.2	84.0±6.6	97.0±9.4	77.0±18.2	72.7±16.2	100±0	99.0±3.1	98.7±1.2
2	100±0	53.0±20.5	48.1±9.3	92.0±19.3	50.0±21.0	51.3±13.4	97.0±4.8	61.0±26.4	52.4±4.6	99.0±3.1	44.0±19.5	49.4±9.7	100±0	66.0±23.6	60.6±7.8
3	100±0	43.0±21.6	33.9±7.6	88.0±14.7	45.0±25.9	41.8±9.6	96.0±5.1	52.0±20.4	51.3±10.5	99.0±3.1	36.0±18.9	33.1±8.1	100±0	63.0±16.3	65.8±12.4
4	100±0	34.0±17.7	38.6±6.0	99.0±3.1	63.0±14.9	58.1±8.7	98.0±4.2	65.0±15.8	65.0±8.7	98.0±4.2	42.0±20.4	42.2±9.4	100±0	91.0±8.7	89.9±7.6
5	100±0	62.0±14.7	61.1±5.4	98.0±6.3	73.0±19.4	74.8±10.8	100±0	75.0±19.0	73.4±7.6	97.0±6.7	56.0±23.1	59.1±10.6	100±0	82.0±15.4	83.3±4.3
6	100±0	84.0±12.6	85.4±4.6	100±0	80.0±17.6	72.2±9.8	100±0	76.0±21.7	74.6±12.2	100±0	97.0±4.8	94.7±2.2	100±0	82.0±12.2	79.9±6.7
7	100±0	93.0±8.2	88.1±5.6	99.0±3.1	94.0±5.1	90.9±5.1	100±0	98.0±4.2	94.3±5.0	100±0	89.0±8.7	83.4±9.6	100±0	94.0±6.9	93.1±5.2
8	100±0	87.0±14.1	77.1±9.7	98.0±4.2	86.0±8.4	75.8±8.0	100±0	90.0±8.1	83.7±4.6	100±0	92.0±9.1	88.9±9.6	100±0	100±0	99.7±0.2
9	100±0	91.0±9.9	85.5±13.2	100±0	97.0±4.8	93.5±9.1	100±0	96.0±5.1	96.0±6.5	100±0	94.0±5.1	89.5±7.2	100±0	100±0	100±0
10	100±0	41.0±17.2	38.1±11.8	94.0±8.4	53.0 ±19.4	49.5±10.2	98.0±4.2	56.0±17.7	49.4±14.3	100±0	41.0±26.8	46.5±15.6	100±0	72.0±19.3	72.3±6.4
11	100±0	35.0±19.5	33.5±4.3	95.0±7.0	60.0±13.3	46.1±10.1	95.0±7.0	54.0±18.3	47.0±8.6	100±0	30.0±13.3	31.2±7.3	100±0	68.0±16.1	75.2±5.2
12	100±0	30.0±22.6	39.4±6.1	91.0±18.5	35.0±18.4	39.3±8.5	98.0±4.2	27.0±23.1	37.2±6.6	97.0±4.8	41.0±25.5	43.5±9.3	100±0	67.0±17.6	72.5±6.5
13	100±0	92.0±9.1	95.2±5.4	100±0	94.0±5.1	96.0±2.3	100.0±0	96.0±5.1	96.6±2.6	100.0±0	100±0	100±0	100±0	98.0±4.2	99.7±0.2
14	100±0	71.0±22.8	69.3±8.2	99.0±3.1	80.0±10.5	74.0±10.9	97.0±4.8	77.0±12.5	75.9±10.0	100.0±0	94.0±6.9	93.5±3.3	100±0	98.0±4.2	94.8±5.8
15	100±0	45.0±23.2	46.3±9.5	94.0±8.4	37.0±12.5	34.6±8.7	100±0	53.0±27.1	56.9±8.9	95.0±7.0	34.0±15.7	31.4±5.9	100±0	96.0±8.4	90.8±10.1
16	100±0	84.0±10.7	86.9±3.3	99.0±3.1	81.0±11.0	88.6±3.3	100±0	92.0±9.1	95.4±4.2	100±0	97.0±4.8	96.4±1.9	100±0	96.0±6.9	96.8±3.3
OA	100±0	63.6±3.2	50.5±2.9	96.6±4.6	69.5±3.6	56.0±3.4	98.6±1.4	71.6±5.0	58.8±2.9	98.8±1.2	66.5±3.3	55.0±1.8	100±0	85.7±3.6	78.0±2.1
AA	100±0	63.6±3.2	62.3±1.9	96.6±4.6	69.5±3.6	66.7±2.7	98.6±1.4	71.6±5.0	70.8±2.2	98.8±1.2	66.5±3.3	66.0±1.4	100±0	85.7±3.6	85.8±1.2
k	100±0	61.2±3.4	44.9±3.1	96.4±4.9	67.4±3.9	50.6±3.8	98.5±1.5	69.8±5.3	53.9±3.2	98.8±1.2	64.2±3.5	49.8±1.9	100±0	84.8±3.8	75.2±2.3

Finally, the overall optimization problem for solving the proposed LRR-PCRC is summarized in Algorithm 1.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Data Sets Description

1) Indian Pines Data Set: The Indian Pines [25] data set has been collected by AVIRIS sensors in June 1992 which consists of urban images. The 20 heavily noise bands have been removed since these bands are severely affected by the water absorptions and thus 200 bands remained. The spectral range varies from 0.2 to 2.4 μm, while the spatial resolution is 20 m per pixel with size of 145 $\times$ 145 pixels. There are in total 10 366 samples within 16 classes need to be classified in this data set.

2) Pavia University Data Set: The Pavia University [25] data set has been collected by ROSIS sensors over the area surrounding the University of Pavia, Italy, in 2001. The 12 heavily noise bands have been removed since these bands are severely affected by the water absorptions and thus 103 bands remained. There are in total 42 776 labeled samples within nine classes available for classification in this data set.

B. Benchmarking Approaches

To validate our proposed LRR-PCRC, we have compared LRR-PCRC with the state-of-the-art methods to benchmark the performance of our method, including the SMLR [9], SVM [34], and SVM with composite kernel (SVM-CK), SMLR with attribute profile (SMLR-AP) [9], PCRC and PCRC-AP. The MATLAB codes of SVM and SVM-CK are downloadable from [35], while the codes of SMLR and AP are downloadable from [36]. The parameters of AP are set according to the recommendation in [9].

The experiments are carried out on a computer with 2.9 GHz i7 7820HQ CPU with 32 GB RAM running win10 OS. The codes are implemented in MATLAB (R2015a) and all the

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TABLE XI

TRAINING, VALIDATION AND TESTING RESULTS IN INDIAN PINES DATA SET (FIVE SAMPLES PER CLASS FOR TRAINING, FIVE SAMPLES PER CLASS FOR VALIDATION, REMAINING FOR TESTING)

	SMLR			SVM			SVM-CK			PCRC			LRR-PCRC		
No.	Tr	Val	Ts	Tr	Val	Ts	Tr	Val	Ts	Tr	Val	Ts	Tr	Val	Ts
1	96.0±6.9	48.0±21.4	37.7±9.0	82.0±13.9	66.0±18.9	55.1±3.1	94.0±13.4	66.0±20.6	60.5±8.6	76.0±12.6	33.0±20.0	21.7±7.3	99.0±3.1	72.0±15.4	59.0±5.1
2	99.0±3.1	62.0±26.9	57.9±10.7	83.0±22.6	62.0±19.8	54.8±14.5	97.0±6.7	68.0±19.3	65.8±12.5	90.0±9.4	69.0±26.0	69.4±12.9	100±0	80.0±18.2	75.6±10.6
3	99.0±3.1	53.0±13.3	61.1±5.4	68.0±27.4	43.0±23.5	48.3±15.5	93.0±15.6	55.0±25.0	64.6±12.0	95.0±5.2	71.0±15.2	78.7±9.2	98.0±4.2	97.0±4.8	94.6±3.9
4	100± 0	86.0±11.7	88.8±5.9	98.0±4.2	82.0±18.1	82.6±11.0	100± 0	76.0±17.7	79.1±10.9	100±0	94.0±8.4	95.3±3.4	100±0	73.0±8.2	77.9±9.4
5	100±0	100± 0	98.9±0.1	100±0	98.0±6.3	96.8±4.0	100±0	100±0	98.6±2.7	100±0	100±0	99.9±0.1	100±0	100±0	99.2±0.3
6	100±0	45.0±19.0	45.6± 5.5	91.0±12.8	43.0±22.1	53.0±12.0	98.0±6.3	49.0±21.8	60.2±9.1	78.0±9.1	24.0±12.6	30.4±9.0	100±0	97.0±4.8	94.0±3.5
7	99.0±3.1	70.0±13.3	59.6±11.2	99.0±3.1	94.0±10.7	91.3±2.4	99.0±3.1	87.0±12.5	86.6±5.8	95.0±7.0	68.0±15.4	59.7±10.2	100±0	99.0±3.1	99.4±0.4
8	100±0	40.0±15.6	48.6±6.7	81.0± 17.2	63.0±25.8	64.3±12.2	95.0±8.4	65.0±20.6	68.1±8.4	90.0±10.5	33.0±15.6	38.2±9.9	98.0±4.2	81.0±12.8	77.6±13.7
9	96.0±5.1	85.0±9.7	85.6±8.1	100±0	99.0±3.1	99.8±0.1	100± 0	91.0±11.0	91.8±6.3	68.0±18.7	49.0±23.3	45.5±17.1	100±0	67.0±22.6	73.8±9.8
OA	98.7±1.1	65.4±6.6	56.8±4.6	89.1±6.7	72.2±6.0	60.5±6.1	97.3±4.8	73±6.3	67.6±6.4	88.0±2.3	60.1±6.2	57.1±5.0	99.4±0.5	85.1±2.7	77.9±3.8
AA	98.7±1.1	65.4± 6.6	64.9±1.8	89.1±6.7	72.2±6.0	71.8±1.7	97.3±4.8	73±6.3	75.0±2.8	88.0±2.3	60.1±6.2	59.9±1.5	99.4±0.5	85.1±2.7	83.5±2.3
k	98.6±1.2	61.1±7.4	46.7±4.6	87.7± 7.6	68.7±6.7	51.5±5.9	97±5.4	69.6±7.0	59.4±7.2	86.5±2.6	55.1±7.0	46.0±4.7	99.3±0.6	83.2±3.0	72.3±4.2

TABLE XII

TIME CONSUMING COMPARISON IN INDIAN PINES AND PAVIA UNIVERSITY DATA SET (20 TRAINING SAMPLES PER CLASS)

Indian Pines	20	SMLR	SVM	SMLR-AP	SVM-AP	SVM-CK	PCRC	PCRC-AP	LRR-PCRC
	s	-	-	AP:1.6	AP:1.6	-	-	AP:1.6	-
	Tr (s)	0.14	12.82	0.05	4.02	26.59	27.24	24.82	18.67
	Ts(s)	0.03	0.16	0.01	0.14	0.21	4.25	2.51	0.39
Pavia University	20	SMLR	SVM	SMLR-AP	SVM-AP	SVM-CK	PCRC	PCRC-AP	LRR-PCRC
	s	-	-	AP:4.3	AP:4.3	-	-	AP: 4.3	-
	Tr (s)	0.06	2.41	0.03	1.37	11.86	40.49	40.75	36.96
	Ts(s)	0.05	0.28	0.03	0.28	0.47	11.51	10.48	0.49

experimental results in this article are the averaged values by repeating 10 times.

The key performance indicators (KPI) include overall accuracies (OA), average accuracies (AA), kappa coefficient (k), category accuracies (CA), training time (Tr), and testing time (Ts). In addition, all the abbreviations in this article have been listed in Table I.

C. Parameter Analysis

The key parameters for PCRC are β and λ in (9) and additional parameters for proposed LRR-PCRC are f in (15) and γ in (14). Four experiments are carried out to evaluate the parameters of λ, β, γ, and f, respectively, using five training samples per class. Since the proposed LRR-PCRC focuses on the feature extraction and classification, we consider two different situations of the raw data, i.e., without additional noise and with normally distribution noise (*i.i.d.:* zero mean with σ^2 covariance). In this section, we set σ to be 0.02. We fix the other parameters when evaluating one parameter. The experimental parameters setting for this section can be seen in Table II. In order to achieve a reliable result, we have iterated each experiment ten times and averaged the results.

Experiment 1 (λ) : In this experiment, we evaluate the parameter $\lambda = 2^{a1}$ which $a1$ ranges from [−20, −19,...,5]. $\beta = 2^{-10}$ for both PCRC and LRR-PCRC. $f = 3$ and $\gamma = 2^{12}$ for LRR-PCRC. It can be seen from Fig. 2(a) and (b) that λ has some impact on both PCRC and LRR-PCRC. In addition, in Indian Pines and Pavia University data sets, the OA of the proposed LRR-PCRC is higher than PCRC under the situation with or without addition noise. We also can see that the OA of PCRC decreased dramatically when add the noise to HSIs data. However, the proposed LRR-PCRC is more robust to noise than PCRC.

TABLE XIII

CLASSIFICATION RESULTS IN INDIAN PINES (20 TRAINING SAMPLES PER CLASS) (BEST RESULTS IN BOLD)

			RoRF-KPCA			
NO.	RF	RoF	Linear	RBF	Poly	LRRPCRC
1	84.81	75.29	89.63	92.22	91.30	**99.7±0.9**
2	32.57	46.55	58.65	57.03	61.93	**86.7±5.8**
3	42.01	50.31	51.97	51.43	51.43	**80.5±4.8**
4	53.12	66.96	83.12	85.17	82.82	**96.5± 3.2**
5	76.66	76.79	86.38	84.85	85.01	**89.6±4.5**
6	67.95	75.25	91.34	92.36	92.01	**96.3±2.9**
7	98.08	78.33	**99.23**	**99.23**	**99.23**	96.9±3.9
8	80.57	78.83	92.70	92.68	93.62	**99.8±0.1**
9	90.00	78.00	98.00	99.00	98.00	**99.0±3.1**
10	48.61	58.39	76.03	76.17	73.85	**84.7±4.4**
11	44.04	37.71	53.16	54.57	52.07	**89.2±2.4**
12	42.56	47.86	80.70	79.06	81.95	**91.2±6.0**
13	93.73	92.97	99.06	98.87	98.73	**99.8±0.2**
14	80.00	81.03	89.10	89.46	91.11	**99.5±0.9**
15	46.08	44.08	62.87	62.47	61.40	**98.9±1.1**
16	96.42	95.33	97.05	97.05	**98.32**	97.8±1.9
OA	54.34	57.02	71.05	71.12	71.30	**91.0±0.8**
AA	67.33	67.73	81.81	81.98	82.04	**94.1± 0.5**
k	49.06	51.99	67.61	67.65	67.85	**89.7±0.9**

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TABLE XIV
CLASSIFICATION RESULTS IN PAVIA UNIVERSITY (20 TRAINING SAMPLES PER CLASS) (BEST RESULTS IN BOLD)

			RoRF-KPCA			
NO.	RF	RoF	Linear	RBF	Poly	LRRPCRC
1	65.56	72.45	**79.01**	75.72	76.05	75.8±3.7
2	**99.99**	99.98	99.97	99.97	99.99	90.9±3.3
3	99.15	**99.56**	99.37	99.12	99.33	98.4±1.6
4	65.81	81.88	78.27	77.69	77.43	**88.1±8.2**
5	82.25	90.67	89.88	89.93	91.95	**99.5±0.2**
6	56.11	70.80	72.91	72.90	74.75	**98.1±2.6**
7	54.02	60.30	58.33	66.88	68.08	**99.6±0.3**
8	68.35	68.96	70.55	70.72	71.12	**92.9±2.3**
9	84.51	82.83	90.89	88.18	90.72	**98.1±3.4**
OA	64.89	74.77	76.16	76.17	77.32	**90.4±1.2**
AA	75.08	80.83	82.13	82.35	83.27	**93.5±0.6**
k	56.61	68.31	69.79	69.79	71.16	**87.5± 1.5**

As shown in Fig. 2(a) and (b), the PCRC achieved best accuracies at about $\lambda = 2^{-7}$ and $\lambda = 2^{-2}$ under the situation with and without noise, respectively, in Indian Pines. In Pavia University data set, the PCRC achieves good classification accuracies at $\lambda = 2^{-7}$ for both no additional noise and noise condition. Hence, above parameters have been selected for PCRC if not otherwise mentioned. For the proposed LRR-PCRC, the classification accuracies are relatively stable ranging from $\lambda = 2^{-15}$ to $\lambda = 2^{-5}$ in Indian Pines data set while ranging from $\lambda = 2^{-15}$ to $\lambda = 2^{3}$ in Pavia University data set. Hence, $\lambda = 2^{-10}$ and $\lambda = 2^{0}$ have been selected for Indian Pines and Pavia University data sets, respectively.

Experiment 2 (β) : In this experiment, we evaluated the parameter $\beta = 2^{a2}$ which $a2$ ranges from [−20, −19, …,5]. It can be seen from Fig. 2(c) and (d) that the β also has a certain impact on PCRC in both Indian Pines and Pavia University data sets, and on LRR-PCRC in Indian Pines data set. In addition, the proposed LRR-PCRC not only produces higher classification accuracies than PCRC, but also demonstrates more robustness to noise than PCRC. As also can be seen from Fig. 2(c) and (d) that about at the value of $\beta = 2^{-10}$, PCRC has produced the best classification accuracies in Indian Pines data set. In Pavia University data set, PCRC produced the best classification accuracies at $\beta = 2^{-7}$ and $\beta = 2^{-4}$ under the condition with and without additional noise, respectively. Hence, above parameters have been selected for PCRC if not otherwise mentioned. For the proposed LRR-PCRC, it performs stably when β varies from [−20,…, −5] and [−20,…,5] in Indian Pines data set and Pavia University data set, respectively. Hence, if not specially mentioned in the following experiments, β is set to 2^{-8} and 2^{3} for the proposed LRR-PCRC in Indian Pines data set and Pavia University data set, respectively.

Experiment 3 (γ) : In this experiment, we evaluated the parameter $\gamma = 2^{a3}$ which $a3$ ranges from [1, 2,…,15]. It can be seen from Fig. 2(e) and (f) that the OA of LRR-PCRC increased then decreased when γ increases. In addition, the noise has more impact on Indian Pines data set than in Pavia University data set when γ is small. However, the impact can be mostly eliminated when γ is set to a higher value. It would be good for classification performance when λ and β are set to small values while γ is set to a big value. This is because the CI may dominate the important role for feature extraction and classification. We can see that LRR-PCRC yield best classification accuracies both in Indian Pines data set and Pavia University data set at $\gamma = 2^{12}$ and $\gamma = 2^{9}$, respectively. Hence, $\gamma = 2^{12}$ and $\gamma = 2^{9}$ have been selected for LRR-PCRC.

Experiment 4 (f) : In this experiment, we evaluated the parameter f which f ranges from [1, 2,…,10]. It can be seen that from Fig. 2(g) and (h) that the OA of LRR-PCRC increases then decreases when f increased. It can be also seen that the best value of f for classification accuracies is 3 in both data set and both situations that with and without additional noise. Therefore, $f = 3$ has been selected in the following experiments.

D. Contribution Analysis

In this section, we analyze the three contributions of the proposed LRR-PCRC. Compared with PCRC, the proposed LRR-PCRC has three main contributions: 1) the LRR-PCRC with sparse representation for improving the feature extraction performance; 2) LRR-PCRC with SR and ED; and 3) LRR-PCRC with SR, ED and CI for extracting the efficient features of HSIs in order to further improve the classification accuracies. Hence, we will show the impact of each contribution of the proposed LRR-PCRC in this section. In order to show the exact impact of contributions on proposed model, we selected the best parameters of these contributions which can be seen in Table III. We used 20 training samples per class under the condition with and without additional noise $\sigma = 0.02$ (i.i.d: zero mean with σ^2 covariance).

As can be seen from Table IV, three contributions have its improvement of classification accuracies in both Indian Pines data set and Pavia University data set. We can also see a phenomenon, ED information seems to be less important in Indian Pines data set but important in Pavia University data set. This can be explained by the reason that the Indian Pines and Pavia University data set have different spatial structures. In more details, as can be seen from Fig. 3, the pixel of the same category in Indian Pines data set are located together, and the shapes of the same category are more likely to be blocked. While in Pavia University data set, the shapes of some categories are more likely to be stripe; hence, the CI has a dominant role in Indian Pines data set. However, in Pavia University data set, both the ED and CI are very important. From the above discussion, it can be seen that it is more reasonable to use CI and ED collaboratively.

E. Effect of Varying Numbers of Training Samples and the Varying Levels of Noise

In this section, we will further evaluate the performance in feature extraction and classification of the proposed LRR-PCRC via varying the numbers of training sample and levels of noise. The training samples vary between 20, 25, and 30, while the noise levels vary between 0.2, 0.4, 0.6 and 0.8.

As can be seen from Tables V and VI, the classification results in Indian Pines data set and Pavia University data set by LRR-PCRC are more stable and better than PCRC when we

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TABLE XV
CONFUSION MATRIX OF PROPOSED LRR-PCRC IN INDIAN PAVIA (20 TRAINING SAMPLES PER CLASS)

		Predict Value															
	Category	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Actual Value	1	33.9	0	0	0	0	0	0.1	0	0	0	0	0	0	0	0	0
	2	10.1	122.61	24.3	3.5	1.8	1.5	0	0	0	64.1	46.3	26.7	0	3.0	5	1.6
	3	0	17.9	655.5	65.8	2.9	0	0	0	5.1	13.8	5.0	46.2	0	0	1.8	0
	4	0	3.4	1.7	206.7	0	1.2	0	0	0	0	0	0.7	0	0	0.3	0
	5	2.4	0	3.6	0	427.4	0	31.9	0	0	0.9	1.4	0.7	2	0	6.7	0
	6	0	0.1	0.1	0	6.3	700.4	0	0	10.0	0	0.2	2.8	0	7.1	0	0
	7	0.2	0	0	0	0.2	0	12.6	0	0	0	0	0	0	0	0	0
	8	0	0	0	0	0	0	0	468.5	0	0	0	0	0	0.5	0	0
	9	0	0	0	0	0	0.1	0	0	9.9	0	0	0	0	0	0	0
	10	1.7	41.3	61.9	0	1	0	0	3.0	0	803.3	13.0	22.3	0.2	0.1	0.2	0
	11	0	170.1	6.8	1.3	4.5	12.2	0	11.8	3.5	46.1	2184.5	0.2	0	0	7	0
	12	0	3.8	18.8	10.7	0.6	0	0	0	0	9.3	0	542.2	0	0	0.4	8.2
	13	0	0	0.2	0	0	0	0	0	0	0	0	0	191.8	0	0	0
	14	0	0	0	0	0	1.6	0	2.9	0	0	0.3	0	0	1268.8	0.4	0
	15	0	0.4	0	0	0	0	0	0	0	0	0.1	2.7	0	0	356.3	0.5
	16	0	0	0	0	0	0	0	0	0	0	0	1.6	0	0	0	73.4

TABLE XVI
CONFUSION MATRIX OF THE PROPOSED LRR-PCRC IN PAVIA UNIVERSITY (20 TRAINING SAMPLES PER CLASS)

		Predict Value								
	Category	1	2	3	4	5	6	7	8	9
Actual Value	1	5011.4	78.1	171.9	0.1	36.7	55.9	649.2	607.7	0
	2	1.9	16941	1.5	1154.3	0	130	0	400.7	0
	3	2	0.2	2046.4	0.6	0	0.8	1.5	27.5	0
	4	0.6	319.7	0.2	2683.8	6	33.1	0.1	0.5	0
	5	0.4	0.9	0	0	1318.4	0.4	0	4.9	0
	6	1.3	0.4	0	4.4	45.3	4915.7	6.1	35.8	0
	7	1.9	0	0.4	0	0	1.6	1305.6	0.5	0
	8	46.8	44.3	137.6	0	0	1.1	26.6	3405.6	0
	9	17.2	0	0	0	0.2	0	0.1	0	909.5

vary the number of training samples and levels of noise. This further demonstrates the superior performance of the proposed LRR-PCRC.

F. Comparison With Other State-of-the-Art Algorithms

In this section, first, we verified the proposed methods using under 1% training samples, and the corresponding training samples and testing samples are shown in Table VII. The experiments are conducted in two conditions, i.e., no additional noise and noise with level of $\sigma = 0.02$. Tables VIII and IX display the classification accuracies in Indian Pines and Pavia University data set. As can be seen from Tables VIII and IX, the classification accuracies of the proposed LRR-PCRC are better than PCRC and other methods in the two conditions. This verifies the good performance of the proposed LRR-PCRC again. Figs. 4 and 5 show the corresponding classification map.

Second, to further verify the proposed methods, we have divided the entire hyperspectral data set into three subsets (denoted as Subset 1, 2, and 3Â'jrandomly ten times. In each time, Subset 1 and Subset 2 are composed of randomly selected samples containing five samples per class. It is used for twofold cross validations, i.e., Subset 1 for training and Subset 2 for validation and vice versa. Subset 3 containing the rest of the samples is used for testing. Finally, all the classification results will be averaged. Tables X and XI show the results of training (Tr), validation (Val), and testing (Ts). As can be seen from these two tables, in general, the proposed method acquired better results than PCRC and other methods in both data sets.

Third, Table XII shows the time consumed in training and testing with 20 training samples per class without additional noise. It should be noted that the time costs of these methods would not be changed since the dimensionality of HSIs has not changed when noise are added to HSIs. From Table XII, we can see that in both Indian Pines and Pavia University data set, the time consumed by our proposed LRR-PCRC is higher than SMLR, SVM, SMLR-AP, and SVM-AP, but that of the proposed LRR-PCRC is less than PCRC and PCRC-AP. The proposed LRR-PCRC has higher time cost than SVM-CK in Pavia University data set but lower in Indian Pines and Pavia University data set. Here, we theoretically

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Algorithm 1 The Proposed LRR-PCRC

Input: For N training samples from K classes of a hyperspectral image $X = [X_{N_1}, X_{N_2}, \ldots, X_{N_K}] \in R^{d\times N}$ and n testing samples $Y = [Y_1, Y_2, \ldots, Y_n] \in R^{d\times n}$, the parameters $\beta, \lambda, \gamma, f, \tau^0 = 10 * \lambda, Y_1 = 0, Y_2 = 0$.

1.1 Set t = 0

1.2 Update $\boldsymbol{H}$:

$H^{t+1} = \arg\min_H \ {\lambda}/{\tau^t}\left\|H^t\right\|_1 + {1}/{2}\|H^t - A^t + Y_1^t/\tau^t\|_F^2 \Longrightarrow H^{t+1} \leftarrow \max\{0, abs\left(A^t - \frac{Y_1^t}{\tau^t}\right) - (\Gamma \times {\lambda}/{\tau^t})\} \times sign(A^t - Y_1^k/\tau^t)$

1.3 Update $\boldsymbol{J}$:

$J^{t+1} = \min_J \ {\gamma}/{\tau^t}\|C \odot J\|_1 + {1}/{2}\|J^t - A^t + Y_2^t/\tau^t\|_F^2 \Longrightarrow J^{t+1} = \max\left\{0, abs\left(A^t - \frac{Y_2^t}{\tau^t}\right) - \left(\Gamma \times {\gamma}/{\tau^t}\right)\right\} \times sign(A^t - Y_2^k/\tau^t)$

1.4: Update $\mathbf{A}$:

$A^{t+1} = \arg\min_A \ {1}/{2}\|Y - XA^t\|_F^2 + {\beta}/{K}\sum_{k=1}^K \|XA^t - X_{N_k}A^t_{N_k,:}\|_F^2 + {\tau^t}/{2}(\|H^t - A^t + Y_1^t/\tau^t\|_F^2 + \|J^t - A^t + \frac{Y_2^t}{\tau^t}\|_F^2)$

$\Longrightarrow A^{t+1} = \left(X^TX + {\beta}/{K}\sum_{k=1}^K \bar{\bar{X}}_{N_k}^T \bar{\bar{X}}_{N_k}) + 2\tau^t I\right)^{-1}(X^TY + \tau^t\left(H^t + J^t\right) + Y_1^t + Y_2^t)$

1.5 Update other parameters:

$Y_1^{t+1} = Y_1^t + \tau^t(H^{t+1} - A^{t+1})$; $Y_2^{t+1} = Y_2^t + \tau^t(J^{t+1} - A^{t+1})$, $\tau^{t+1} = 1.1\times\tau^t$.

1.6 Quit the algorithm if the stopping criterion is met; otherwise, go back to Step 1.2.

Output: Predict the testing sample label:

$la_{Y_i^*} = \arg\max_k \ \exp(-\|X\hat{\alpha} - X_{N_k}\hat{\alpha}_{N_k}\|_2^2)$ or $la_{Y_i^*} = \arg\max_k \ (\|X\hat{\alpha} - X_{N_k}\hat{\alpha}_{N_k}\|_2^2)$; $i = 1, \ldots, n$, $k = 1, \ldots, \mathrm{K}$

analyze the computation complexity of the proposed LRR-PCRC to reveal the reason behind the scene. As one can see from Algorithm 1, the proposed LRR-PCRC has four updating steps (Step 1.2–Step 1.5). The major cost is the inverse operation in Step 1.4 and this is the main reason that the proposed method has high computational time compared with SVM-based and SMLR-based methods. The proposed LRR-PCRC needs to compute the inverse operation in each iteration. The computation cost of matrix multiplication and addition in Step 1.2, 1.3, and 1.5 are negligible comparing to the inverse operation. Given a square matrix of the size $g \times g$, the computation complexity of inverse operation for such matrix is $O(g^3)$ and thus the computation complexity of our method is $O(N^3)$. Furthermore, we can see that the classification accuracies of the proposed LRR-PCRC are much higher than the other classifiers. Hence, we can conclude that the proposed LRR-PCRC has good performance in terms of above analysis.

G. Extended Experiments and Analysis

In this section, in order to show the performance of the proposed LRR-PCRC, we have conducted more comparison experiments with the well-known method random forest (RF) [37] and its variations, including rotation forest (RoF) [38] and rotation random forest-kernel principal component analysis (RoRF-KPCA) [39].

The experiments are conducted under the situation that 20 samples per class are used for training and the remaining is used for testing. The kernel function of RoRF-KPCA has three different types, including linear function, radial basis function (RBF), and polynominal (Poly) function. The results of RF, RoF and RoRF-KPCA are directly taken from [39]. Tables XIII and XIV show the classification results of the proposed LRR-PCRC, the RF and its variations in Indian Pines and Pavia University data sets, respectively. As can be seen from these two tables, in general, the proposed method has obtained better classification accuracies than RF and its variations. In addition, the confusion matrices which are corresponding to the classification results of the proposed method in both data sets have been shown in Tables XV and XVI. We, hence, can conclude that the proposed method has achieved good performance.

V. Conclusion

In this article, a novel framework based on sparse constraint and prior information of HSIs, LRR-PCRC, has been proposed to extract efficient feature and classify HSIs. By imposing the sparse constraint to PCRC, the proposed RPCRC can tolerate environmental noise and thus extracts efficient features of HSIs and improves the classification accuracies. In addition, by adding the prior information of HSIs (ED and CI) to RPCRC, the proposed LRR-PCRC can improve the classification accuracies significantly. Experiments have been conducted to compare our proposed LRR-PCRC with other state-of-the-art methods, and the results show our methods have superior performances.

Currently, the computational time of our algorithm is relatively higher compared to SVM or SMLR due to the inverse operation. One direction of our further work will be focused on time costs reduction. One promising way is to use mathematical models to find theoretical solutions; the other approach is seeking dimensionality reductions methods or semisupervised learning practically to further improve the classification accuracies and in the meantime reduce the computational time costs. With respect to the prerequisite of the proposed method, we need to acquire labeled training samples for training the model due to the supervised learning nature of the method. Hence, another research stream of our work will extend the method to unsupervised learning.

Appendix

A. Solution of LRR-PCRC Without ED Nor CI

First, we introduce the auxiliary variable H to split the variable to simplify the problem of (12). Thus, (12) can be

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converted to

$$\hat{A} = \arg\min_{A} \left\{ 1/2\|Y - XA\|_F^2 + \lambda\|H\|_1 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k} A_{N_k,:}\|_F^2 \right\} \quad \text{s.t. } H = A. \tag{24}$$

Then, the corresponding augmented Lagrangian function for (24) can be rewritten as

$$\min_{A,H,J} \left\{ 1/2\|Y - XA\|_F^2 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k} A_{N_k,:}\|_F^2 + \lambda\|H\|_1 + \langle Y_1, H - A\rangle + \tau/2\|H - A\|_F^2 \right\} \tag{25}$$

where $\langle Y_1, H - A\rangle = \text{trace}(Y_1^T(H - A))$, $\tau > 0$ is a penalty parameter. Y_1 is the Lagrange multipliers. Then the alternative optimization algorithm [26] can be applied to solve the model of (25). The details can be seen as follows.

Update H: Fix A, then the H can be updated as

$$H^{t+1} = \arg\min_{H} \lambda/\tau^t \|H^t\|_1 + 1/2\|H^t - A^t + Y_1^t/\tau^t\|_F^2. \tag{26}$$

The solution of (26) can be solved by the simple soft-threshold [32]

$$H^{t+1} = \text{soft}(A^t - Y_1^t/\tau^t, \lambda/\tau^t) = \max\{0, \text{abs}(e) - \lambda/\tau^t \times \text{sign}(e) \tag{27}$$

where abs(e) is the absolute value of e, $e = A^t - Y_1^k/\tau^t$, and sign is the sign function [33].

Update A : Fix H, then the A can be updated as

$$A^{t+1} = \min_{A} 1/2\|Y - XA^t\|_F^2 + \beta/K \sum_{k=1}^{K} \|XA^t - X_{N_k} A_{N_k,:}^t\|_F^2 + \tau^t/2\|H^t - A^t + Y_1^t/\tau^t\|_F^2. \tag{28}$$

Then the solution of (28) can be achieved by the first-order derivation

$$A^{t+1} = \left(X^T X + + \beta/K \sum_{k=1}^{K} \bar{\bar{X}}_{N_k}^T \bar{\bar{X}}_{N_k}) + \tau^t I \right)^{-1} (X^T Y + \tau^t H^t + Y_1^t). \tag{29}$$

B. Solution of LRR-PCRC Without ED or CI

The corresponding augmented Lagrangian function for LRR-PCRC without ED or CI can be written as follows.

1) LRR-PCRC Without CI:

$$\min_{A,H,J} \left\{ 1/2\|Y - XA\|_F^2 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k} A_{N_k,:}\|_F^2 \right\} + \lambda\|\Gamma \odot H\|_1 + \langle Y_1, H - A\rangle + \tau/2\|H - A\|_F^2. \tag{30}$$

Update H: Fix A, then the H can be updated as

$$H^{t+1} = \arg\min_{H} \lambda/\tau^t \|\Gamma \odot H^t\|_1 + 1/2\|H^t - A^t + Y_1^t/\tau^t\|_F^2. \tag{31}$$

The solution of (31) can be solved by the transformation of the soft-threshold rule

$$H^{t+1} = \text{soft}\left(A^t - \frac{Y_1^t}{\tau^t}, \Gamma \times \lambda/\tau^t\right) = \max\left\{0, \text{abs}(e) - \Gamma \times \lambda/\tau^t\right\} \times \text{sign}(e) \tag{32}$$

where $e = A^t - Y_1^t/\tau^t$.

Update A: Fix H, then A can be updated as

$$A^{t+1} = \arg\min_{A} 1/2\|Y - XA^t\|_F^2 + \beta/K \sum_{k=1}^{K} \|XA^t - X_{N_k} A_{N_k,:}^t\|_F^2 + \tau^t/2\|H^t - A^t + Y_1^t/\tau^t\|_F^2. \tag{33}$$

Then the solution of (33) can be achieved by the first-order derivation

$$A^{t+1} = \left(X^T X + \beta/K \sum_{k=1}^{K} \bar{\bar{X}}_{N_k}^T \bar{\bar{X}}_{N_k}) + \tau^t I \right)^{-1} \times (X^T Y + \tau^t H^t + Y_1^t). \tag{34}$$

2) LRR-PCRC Without ED:

$$\min_{A,H,J} \left\{ 1/2\|Y - XA\|_F^2 + \beta/K \sum_{k=1}^{K} \|XA - X_{N_k} A_{N_k,:}\|_F^2 + \gamma\|C \odot J\|_1 \right\} + \langle Y_2, J - A\rangle + \tau/2\|J - A\|_F^2 \tag{35}$$

where J is auxiliary variable to split the variable to let the model become more easily solvable, Y_2 is the Lagrange multipliers.

Update J : Fix A, then J can be updated as

$$J^{t+1} = \arg\min_{H} \gamma/\tau^t \|C \odot J\|_1 + 1/2\|J^t - A^t + Y_2^t/\tau^t\|_F^2. \tag{36}$$

The solution of (36) can be solved by the transformation of the soft-threshold rule

$$J^{t+1} = \text{soft}\left(A^t - \frac{Y_2^t}{\tau^t}, C \times \lambda/\tau^t\right) = \max\left\{0, \text{abs}(e) - \left(\Gamma \times \lambda/\tau^t\right)\right\} \times \text{sign}(e) \tag{37}$$

where $e = A^t - Y_2^t/\tau^t$.

Update A : Fix J, then the A can be updated as

$$A^{t+1} = \arg\min_{A} 1/2\|Y - XA^t\|_F^2 + \beta/K \times \sum_{k=1}^{K} \|XA^t - X_{N_k} A_{N_k,:}^t\|_F^2 + \tau^t/2\|J^t - A^t + Y_2^t/\tau^t\|_F^2. \tag{38}$$

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Then the solution of (38) can be achieved by the first-order derivation

$$A^{t+1} = \left(X^T X + +^{\beta}/_K \sum_{k=1}^{K} \bar{\bar{X}}_{N_k}^T \bar{\bar{X}}_{N_k}) + \tau^t I \right)^{-1} \times \left(X^T Y + \tau^t J^t + Y_2^t \right). \quad (39)$$

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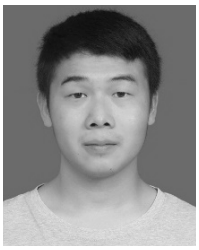
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